1 Kernel and Injectivity

Definition 1.1 (Kernel)**.** The *kernel* of a linear map $T : V \to W$ is the set of vectors in V which T maps to the zero vector of W:

$$\ker T := \{v \in V : Tv = 0\} \subseteq V.$$

Exercise 1.2. Let $T \in \mathscr{L}(V,W)$. Show that $\ker T$ is a subspace of V.

Exercise 1.3. Find the kernel for each of the following linear maps.

(a) The zero map $0 : V \to W$.

(b) $T : \mathbb{C}^3 \to \mathbb{C}$ defined by $T(z_1,z_2,z_3) = z_1 + 2z_2 + 3z_3$. (Give a basis of $\ker T$.)

(c) Let $V = \mathscr{P}(\mathbb{R})$ be the vector space of all polynomials of a real variable with real coefficients, and $D : \mathscr{P}(\mathbb{R}) \to \mathscr{P}(\mathbb{R})$ differentiation.

(d) $T : \mathscr{P}(\mathbb{R}) \to \mathscr{P}(\mathbb{R})$ is the "multiplication by x^2" map, defined by $(Tp)(x) = x^2 p(x)$.

(e) Let $T : F^\omega \to F^\omega$ be the "backward shift" map defined by $T(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots)$.

Definition 1.4 (Injective)**.** A map of sets $f : A \to B$ is *injective* or *one-to-one* if $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$. Or, equivalently, $f(a_1) = f(a_2)$ implies $a_1 = a_2$ (this is just the contrapositive of the first statement).

Proposition 1.5 (An injective linear map has trivial kernel)**.** *A linear map $T : V \to W$ is injective if and only if* $\ker T = \{0\}$.

Exercise 1.6. Prove Proposition 1.5.

Exercise 1.7. Which linear maps in Exercise 1.3 are injective?

2 Image and Surjectivity

Definition 2.1 (Image)**.** Let $T : V \to W$ be a linear map. The *image* (or *range*) of T is the set of all outputs of T:

$$\text{im}T := \{Tv : v \in V\} \subseteq W.$$

Exercise 2.2. Show that im$T$ is a subspace of W.

Exercise 2.3. Determine the image of each of the following linear maps.

(a) The zero map $0 : V \to W$.

(b) Define $T : \mathbb{R}^2 \to \mathbb{R}^3$ by $T(x,y) = (2x, 5y, x+y)$. Given a basis of im$T$.

(c) The differentiation map $D : \mathscr{P}(\mathbb{R}) \to \mathscr{P}(\mathbb{R})$.

Definition 2.4 (Surjective)**.** A map of sets $f : A \to B$ is *surjective* if $\mathrm{im} f = B$. That is, if for every $b \in B$ there exists at least one $a \in A$ such that $b = f(a)$. If f is surjective, one also says that f *maps* A *onto* B.

Exercise 2.5. Which of the linear maps in Exercise 2.3 are surjective?

Exercise 2.6. If we instead consider the differentiation map D on $\mathscr{P}_5(\mathbb{R})$ (the vector space of polynomials of degree at most 5), is D surjective? This shows that surjectivity of a map $T : V \to W$ depends on W.